P1

SIMPLE LINEAR MODEL

$$y_i = \beta_0 + \beta_1 \cdot x_i + \varepsilon_i \; ; \; var(x) > 0$$

①

$$y_i = \beta_0 + \beta_1 \cdot x_i + \varepsilon_i$$

$$\Rightarrow \varepsilon_i = y_i - \beta_0 - \beta_1 \cdot x_i$$

$$\min_{\beta_0, \beta_1} \sum_{i=1}^{n} \hat{\varepsilon}_i^2 = \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 \cdot x_i)^2$$

$$\{\hat{\beta}_0\} : 2 \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 \cdot x_i) \cdot (-1) = 0$$

$$= -\sum_{i=1}^{n} y_i + \sum_{i=1}^{n} \hat{\beta}_0 + \hat{\beta}_1 \cdot \sum_{i=1}^{n} x_i = 0$$

$$\therefore n \hat{\beta}_0 = \sum_{i=1}^{n} y_i - \hat{\beta}_1 \cdot \sum_{i=1}^{n} x_i$$

$$\boxed{\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \cdot \bar{x}}$$

WE ARE MINIMIZING:

$$= \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 \cdot x_i)^2$$

WITHOUT REPLACEMENT

$$= \sum_{i=1}^{n} (y_i - \bar{y} + \hat{\beta}_1 \cdot \bar{x} - \hat{\beta}_1 \cdot x_i)^2 \Longleftrightarrow (2) \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) \cdot x_i = 0$$

$$= \sum_{i=1}^{n} (y_i - \bar{y} - \hat{\beta}_1 (x_i - \bar{x}))^2$$

F.O.C:

$$\{\hat{\beta}_1\} : 2 \cdot \sum_{i=1}^{n} (y_i - \bar{y} - \hat{\beta}_1 \cdot (x_i - \bar{x})) \cdot (-1) \cdot (x_i - \bar{x})$$

$$\Rightarrow -\sum_{i=1}^{n} (y_i - \bar{y}) \cdot (x_i - \bar{x}) + \hat{\beta}_1 \cdot \sum_{i=1}^{n} (x_i - \bar{x})^2 = 0$$

$$\boxed{\therefore \hat{\beta}_1 = \frac{\sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}}$$

NOTE: YOU CAN ALSO WRITE $\hat{\beta}_1$ AS:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} (x_i - \bar{x}) \cdot y_i - \sum_{i=1}^{n} (x_i - \bar{x}) \cdot \bar{y}}{\sum_{i=1}^{n} (x_i - \bar{x})^2} \; ,$$

BUT $\sum_{i=1}^{n} (x_i - \bar{x}) \cdot \bar{y} = \bar{y} \sum_{i=1}^{n} (x_i - \bar{x})$

$$\therefore \hat{\beta}_1 = \frac{\sum_{i=1}^{n} (x_i - \bar{x}) \cdot y_i}{\sum_{i=1}^{n} (x_i - \bar{x})^2}$$

AND

$$\sum_{i=1}^{n} (x_i - \bar{x}) = \sum_{i=1}^{n} x_i - \sum_{i=1}^{n} \bar{x}$$

$$= \sum_{i=1}^{n} x_i - n \cdot \bar{x}$$

$$= \sum_{i=1}^{n} x_i - \cancel{n} \frac{1}{\cancel{n}} \sum_{i=1}^{n} x_i$$

$$= 0$$

② ORTHOGONALITY PROPERTIES

① RESIDUALS SUM TO ZERO

FROM (1)

$$= \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 \cdot x_i)$$

$$= \sum_{i=1}^{n} \hat{u}_i = 0$$

② FROM (2): RESIDUALS ARE ORTHOGONAL TO REGRESSORS

$$\hookrightarrow \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 \cdot x_i) \cdot x_i = 0$$

$$\hookrightarrow \sum_{i=1}^{n} \hat{u}_i \cdot x_i = 0$$

③ RESIDUALS ARE ORTHOGONAL TO $(x_i - \bar{x})$

$$\sum_{i=1}^{n} (x_i - \bar{x}) \cdot \hat{u}_i = \underbrace{\sum_{i=1}^{n} x_i \cdot \hat{u}_i}_{\text{by } (2)} - \sum_{i=1}^{n} \bar{x} \cdot \hat{u}_i$$

$$= 0 - \bar{x} \cdot \cancel{\sum_{i=1}^{n} \hat{u}_i} \; 0$$